

Gönül Demir

Senior Electronics / Hardware R&D Engineer

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Date of Birth: 30.04.1984 | Nationality: Turkish



PROFESSIONAL SUMMARY

Electronics engineer with 15+ years of hands-on experience in hardware development, optoelectronics, sensor systems and EMC/EMI. I have designed and improved real products across beam and flame detection, encoders, industrial sensing and embedded control — from circuit design and prototype validation to certification, documentation and production support. My background combines practical R&D execution, multidisciplinary project ownership and strong interest in technical writing and knowledge sharing.

CORE TECHNICAL SKILLS

Optoelectronics & Sensors Beam and flame detection · optical / magnetic encoders · proximity & diffuse sensing · ToF sensing · MEMS tilt · optical transmitter/receiver chains	Hardware, Embedded & Interfaces Analog/digital circuit design · op-amp signal paths · STM32L072 hardware · RS-422 / RS-485 · 5/24 V industrial interfaces · short-circuit protection
EMC, Compliance & Validation EMC/EMI debugging · grounding / shielding / filtering · CE / LVD / EMC awareness · TSE / LPCB documentation support · prototype bring-up · test automation	Tools & Project Delivery Altium Designer · KiCad · LTspice / SPICE · oscilloscope & lab validation · technical documentation · cross-functional project coordination

PROFESSIONAL EXPERIENCE

- Senior R&D Engineer / Project Lead | Teknim Electronic · Istanbul, Türkiye

2022-2024

 - Led the development of a long-range beam smoke-detector project from feasibility, architecture and design reviews through prototype validation and production transfer.
 - Designed analog and digital circuits, optical transmitter/receiver hardware and STM32L072-based electronics; supported BLE-assisted setup and field-oriented product use.
 - Owned component selection, schematic design, PCB review, prototype bring-up, optical alignment and lab troubleshooting; introduced EMC improvements through filtering, grounding and layout changes.
 - Worked closely with software, mechanical, production and quality teams while also contributing to technical documentation, product documentation and qualification / certification activities.
- R&D Engineer / Project Lead | Fenac Encoders & Sensors · Istanbul, Türkiye

2017-2022

 - Led and contributed to multiple R&D projects including optical and magnetic encoders, wheel encoders, elevator door detection, proximity / diffuse sensors, ToF-based escalator sensing and MEMS tilt products.
 - Designed industrial interface circuits including 5/24 V, 200 mA line-driver stages, RS-422 / RS-485 hardware and short-circuit-protected output structures for robust field operation.
 - Performed optical design, hardware development, prototype validation and EMC troubleshooting across sensor products for civilian and defence-oriented applications.
 - Contributed to advanced development topics including FPGA-based laser-distance work, sensor feasibility studies, test activities and technical documentation.
- R&D Engineer | Mavili Elektronik · Istanbul, Türkiye

2011-2016

 - Designed and developed beam smoke detectors, siren / flasher electronics and four flame-detector models up to prototype stage for industrial fire-detection systems.
 - Worked directly on certification and compliance activities, including EMC testing as well as TSE and LPCB-related documentation / validation for multiple beam-detector variants.
 - Built production and validation test equipment, supported internal and external EMC tests and converted test findings into practical circuit and PCB improvements.

- Supported production engineering and quality improvement through custom test boards and fixtures that helped shorten test time and reduce production defects.

Freelance Electronics Engineer | Inovenso · Istanbul, Türkiye

2010–2011

- Designed motor-driver and distance-sensing electronics for early-stage product and laboratory applications.

Purchasing Engineer | Soypak · Istanbul, Türkiye

2009–2010

- Supported technical purchasing and supplier coordination at the start of my engineering career.

TECHNICAL WRITING & OPEN ENGINEERING

- Publishes practical engineering articles and long-form notes on optical sensing system design, EMC / EMI, shielding, cable shield / reference connections, MOSFET gate drive, peak-detector circuits, adaptive optics, NIR sensing and related hardware topics.
- Maintains an engineering website and shares calculations, design trade-offs, figures and test-oriented reasoning through gonuldemir.github.io and LinkedIn.
- Current openly documented project: 1–30 MHz, 1.3 kW solid-state SP2T RF switch, covering SiC MOSFET selection, isolated gate drive, OFF-state capacitance, thermal design and staged RF validation.

EDUCATION

BSc, Electronics Engineering | Uludağ University · Bursa, Türkiye

2009

Erasmus exchange — Electronic Engineering, Università degli Studi di Salerno, Italy

LANGUAGES

Turkish — Native | English — Professional working proficiency | German — A1

PROFESSIONAL DEVELOPMENT

- Fresu Electronics Academy (Dario Fresu): EMC / EMI Design, Power Integrity (PDN) and Signal Integrity.
- Würth Elektronik: Electromagnetic Compatibility Design & Compliance Issues Seminar.
- Ongoing hands-on development in KiCad, EMC-aware PCB design and high-frequency / high-reliability hardware work.